

TS-RAG TIME: completed default comparison

19 September 2026

Main findings

On 90 matched TIME tasks, default TS-RAG has 12.31% higher arithmetic mean task MASE than maximum-context Chronos-2. The frozen validation mixture has 0.24% higher mean MASE. This comparison establishes no aggregate accuracy benefit for retrieval or mixing. TS-RAG uses maximum-context Chronos-Bolt fallback on 42.75% of scored rows. Newly published compact metadata shows the mixture’s validation support, but its aggregate reason counts combine validation and test and are not eligible evaluation evidence.

1 Question and performed protocol

Does retrieval improve a frozen model beyond the effect of context length, and can validation mixing improve the stronger control? For observed history x and future $y \in \mathbb{R}^H$, with horizon H , each method produces a median point forecast $\hat{y}$. The controls are Chronos-T5-base at 512 observations, Chronos-Bolt-base at 512 and 2,048, and Chronos-2 at 8,192. History is capped at these lengths; controls are univariate without covariates. T5 uses 20 forecast samples. Seed zero is used once.

Default TS-RAG retrieves ten neighbors using T5 representations across all items and variates (measured channels) in the same dataset. Complete 512-point histories and 64-point continuations must be observed by the real query date. The released mixture-of-experts adaptation module emits 64-point chunks. Longer horizons roll forward while neighbor eligibility stays anchored to the real query date. Date-phase alignment and query-scale normalization are disabled. Unavailable retrieval or invalid native forecasts use the independent 2,048-context Bolt forecast for the entire window/variate.

Validation precedes test. With w TS-RAG wins in n usable paired validation trials (ties count one half), freeze $p = (1 + w)/(2 + n)$ before test. No usable validation gives $p = 0$. The mixture is $(1 - p)\hat{y}_{\text{Chronos2}} + p\hat{y}_{\text{TS-RAG}}$. Test retrieval admits observed validation history under the same causal rule.

2 Evaluation and accuracy

MASE divides window mean absolute error by mean absolute seasonal difference over finite historical pairs in the observed prefix. Task means average finite item/window/variate values; the table equally averages 90 tasks. Lower is better. All six methods have finite MASE on the same 67,502 declared cells; the report selects 540 completed evaluations.

Method	Mean task MASE	Inference seconds
Chronos-T5, 512	1.497300	4,875.84
Chronos-Bolt, 512	1.213402	84.77
Chronos-Bolt, 2,048	1.167905	251.32
Chronos-2, 8,192	1.069230	236.76
TS-RAG	1.200871	18,115.12
Frozen mixture	1.071780	18,351.90

3 Paired comparisons and fallback

TS-RAG wins eight tasks and loses 82 against Chronos-2. The mixture wins 36 and loses 54. Against Bolt-max, TS-RAG has 2.82% higher mean MASE, with 19 wins, 52 losses and 19 ties. Increasing Bolt’s context alone improves its mean relative to Bolt-512; retrieval does not beat Bolt-max on this mean.

TS-RAG falls back on 28,860 of 67,502 scored rows (42.75%). Fallback occurs in 49 tasks: 19 have complete and 30 partial fallback; 41 have none. Full-fallback task means match Bolt-max. Complete coverage does not establish native TS-RAG success on every row. No task-level errors are reported.

The published aggregate cause counts combine validation and test, although only test is evaluated. They are therefore excluded from scientific conclusions. The corrected artifact contract reports aggregate fallback causes on test alone while retaining validation masks and per-row reasons as separate mixture diagnostics. Test-only cause totals require a new run under that contract.

Frozen TS-RAG mixture weights range from 0.0455 to 0.7143, with mean 0.3454. All 90 weights are estimated from at least one usable validation date. Validation support totals 27,914 dates, ranges from 1 to 14,868 per task, and averages 310.16. The wide support range means task weights have very different estimation precision; the present study does not quantify it.

4 Cost and evidence limits

TS-RAG additionally records 55,578.80 seconds of datastore preprocessing. Its inference total reuses computed Bolt-max fallback forecasts; mixture totals include both components and mixing. These are recorded pipeline costs, not independent fresh-process latency. They support no speed benefit.

The completed report’s selected manifests, compact MASE means and population dispersion agree. The inspected publication now contains all 90 preparation, 90 extraction, 540 prediction and 90 mixture-weight records. Aggregate calibration support extends the mixture diagnosis without changing the accuracy result; the phase-combined fallback causes do not.

Raw predictions, metric arrays, shared Seasonal masks and per-row reason arrays have not been independently audited. Test-only cause totals and native-only accuracy on a common support are not yet reported. One seed provides no across-run uncertainty or significance claim. Population variance/std describes finite metric cells within a task. Conclusions apply only to the default retrieval settings. The unfinished 16-cell retrieval attempt produced no result and is retired because its schedule is too long; a smaller replacement grid remains to be selected.